Slowly Changing Dimension (SCD) Type 1 Example

Maintain audit columns:

- **Load_Date** → When the record was first inserted.
- **Last_Update_Date** → When the record was last updated.

1. **New Key (Primary key Column) (e.g, Customer_ID not in target)** → Insert as a new record.
2. **Existing Key (Patient_ID exists in target)** →
    a. If **any attribute changes** (e.g., address, city, phone), then **overwrite old values** and update **Last_Update_Date**.
    b. If **no change**, then do not update records.
3. **History is NOT maintained** (old values are lost)

Example of Slowly Changing Dimension (SCD) Type 1 with two audit columns: Load Date and Last Update Date.

IntelliBI

Initial Data Load

Customer_ID	Customer_Name	City	Load_Date	Last_Update_Date
1	John Smith	New York	2025-09-01	2025-09-01
2	Alice Brown	Chicago	2025-09-01	2025-09-01
3	David Johnson	Houston	2025-09-01	2025-09-01

Data After Update (SCD Type 1 Overwrite)

Suppose Alice Brown moves from Chicago to Boston. In SCD Type 1, the existing record is overwritten with the new value, and the Last_Update_Date is refreshed.

Customer_ID	Customer_Name	City	Load_Date	Last_Update_Date
1	John Smith	New York	2025-09-01	2025-09-01
2	Alice Brown	Boston	2025-09-01	2025-09-15
3	David Johnson	Houston	2025-09-01	2025-09-01